MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 10 Solutions

Ex. 5.1.58 Transform the following by making the change of variable $j = i - 1$:

$$\prod_{i=n}^{2n} \frac{n-i+1}{n+i}.$$

SOLUTION . We make the change of variable $j = i - 1$.

Lower limit was $i = n$. So $j = i - 1 = n - 1$.

Upper limit was $i = 2n$. So $j = i - 1 = 2n - 1$.

General term was $\frac{n-i+1}{n+i}$ where $i = j + 1$. So, the general term now becomes $\frac{n-(j+1)+1}{n+(j+1)} = \frac{n-j}{n+j+1}$. Finally, the product becomes

$$\prod_{j=n-1}^{2n-1} \frac{n-j}{n+j+1}.$$

□

Ex. 5.2.12 Prove the following by mathematical induction:

$$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all integers $n \geq 1$.

SOLUTION . Let $P(n)$ be the statement $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$.

Base case is $P(1)$ which is the equation $\frac{1}{1\cdot 2} = \frac{1}{1+1}$, which is of course true.

Now assume $P(k)$ holds, for $k \geq 1$. So we assume $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{k(k+1)} = \frac{k}{k+1}$. We now prove $P(k+1)$.

$$\begin{aligned}
\text{LHS of } P(k+1) &= \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\
&= \left(\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{k(k+1)}\right) + \frac{1}{(k+1)(k+2)} \\
&\text{By Inductive Hypothesis } P(k) \\
&= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\
&= \frac{k(k+2)+1}{(k+1)(k+2)} \\
&= \frac{k^2+2k+1}{(k+1)(k+2)} \\
&= \frac{(k+1)^2}{(k+1)(k+2)} \\
&= \frac{k+1}{k+2} \\
&= \text{RHS of } P(k+1).
\end{aligned}$$

$\square$

Ex. 5.3.17 Prove the following by mathematical induction:

$$1 + 3n < 4^n \text{ for every integer } n > 1.$$

SOLUTION . Let $P(n)$ be the statement $1 + 3n < 4^n$.

Base case is $P(2)$ which is the inequality $1 + 3 \cdot 2 < 4^2$ which is the same as $7 < 16$. Now assume that $P(k)$ holds for $k \geq 2$, i.e. assume that $1 + 3k < 4^k$. Now

$$\begin{aligned}
\text{LHS of } P(k+1) &= 1 + 3(k+1) \\
&= 1 + 3k + 3 \\
&\text{By Inductive Hypothesis } P(k) \\
&< 4^k + 3 \\
&< 4^k + 4 \\
&< 4^k + 4^k + 4^k + 4^k \\
&= 4 \cdot 4^k \\
&= 4^{k+1} \\
&= \text{RHS of } P(k+1).
\end{aligned}$$

$\square$

Ex. 6.1.4. Let $A = \{n \in \mathbb{Z} | n = 5r \text{ for some integer } r\}$ and $B = \{m \in \mathbb{Z} | m = 20s \text{ for some integer } s\}$.

a. Is $A \subseteq B$? Explain.
b. Is $B \subseteq A$? Explain.

SOLUTION . a. No. Consider $5 \in \mathbb{Z}$. Since $5 = 5 \cdot 1$, and 1 is an integer, we know that $5 \in A$.
Suppose $5 \in B$. Then $5 = 20s$ for some integer s. Dividing both sides by 20 gives $s = \dfrac{5}{20} = \dfrac{1}{4}$ which is not an integer. This contradicts s is an integer, and so $5 \notin B$.

b. Yes. Let m be an integer in B. Then $m = 20s$ for some integer s.
We may express $m = 5(4s)$ and clearly, $4s$ is also an integer.
Therefore $m \in A$.

$\square$

Ex. 6.1.29. Let $\mathbb{R}$ be the set of all real numbers. Is $\{\mathbb{R}^+, \mathbb{R}^-, \{0\}\}$ a partition of $\mathbb{R}$? Explain your answer.

SOLUTION . Yes. This is because every real number x satisfies exactly one of the conditions: $x > 0$ or $x = 0$ or $x < 0$. $\square$

Ex. 6.1.33.

a. Find $\mathscr{P}(\varnothing)$.
b. Find $\mathscr{P}(\mathscr{P}(\varnothing))$.
c. Find $\mathscr{P}(\mathscr{P}(\mathscr{P}(\varnothing)))$.

SOLUTION . a. $\mathscr{P}(\varnothing) = \{\varnothing\}$.

b. $\mathscr{P}(\mathscr{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\}$.

c. $\mathscr{P}(\mathscr{P}(\mathscr{P}(\varnothing))) = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$.

If you are interested, this is what we get for $\mathscr{P}(\mathscr{P}(\mathscr{P}(\mathscr{P}(\varnothing))))$. Call $A_1 := \varnothing$, $A_2 := \{\varnothing\}$, $A_3 := \{\{\varnothing\}\}$, and $A_4 := \{\varnothing, \{\varnothing\}\}$. Then

$$
\begin{aligned}
&\mathscr{P}(\mathscr{P}(\mathscr{P}(\mathscr{P}(\varnothing))))\\
&= \Big\{ \varnothing,\\
&\quad \{A_1\},\ \{A_2\},\ \{A_3\},\ \{A_4\},\\
&\quad \{A_1, A_2\},\ \{A_1, A_3\},\ \{A_1, A_4\},\ \{A_2, A_3\},\ \{A_2, A_4\}, \{A_3, A_4\},\\
&\quad \{A_1, A_2, A_3\},\ \{A_1, A_2, A_4\},\ \{A_1, A_3, A_4\},\ \{A_2, A_3, A_4\},\\
&\quad \{A_1, A_2, A_3, A_4\}\Big\}\\
&= \Big\{ \varnothing,\\
&\quad \{\varnothing\},\ \{\{\varnothing\}\},\ \{\{\{\varnothing\}\}\},\ \{\{\varnothing,\ \{\varnothing\}\}\},\\
&\quad \{\varnothing, \{\varnothing\}\},\ \{\varnothing, \{\{\varnothing\}\}\},\ \{\varnothing, \{\varnothing, \{\varnothing\}\}\},\ \{\{\varnothing\}, \{\{\varnothing\}\}\},\ \{\{\varnothing\}, \{\varnothing, \{\varnothing\}\}\},\ \{\{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\},\\
&\quad \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\},\ \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\},\ \{\varnothing, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\},\ \{\{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\},\\
&\quad \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}\Big\}.
\end{aligned}
$$

$\square$

Ex. 6.2.19. Use an element argument to prove the following statements. Assume that all sets are subsets of a universal set U.

For all sets A, B, and C,

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

SOLUTION . 1. Show: $A \times (B \cap C) \subseteq (A \times B) \cap (A \times C)$.

Let $(x, y) \in A \times (B \cap C)$. Then $x \in A$ and $y \in B \cap C$ (by definition of Cartesian product). Therefore $x \in A$ and $y \in B$ and $y \in C$ (definition of intersection). From this, we conclude that $(x, y) \in A \times B$ and $(x, y) \in A \times C$ (definition of Cartesian product). Therefore, $(x, y) \in (A \times B) \cap (A \times C)$ (definition of intersection).

2. Show: $(A \times B) \cap (A \times C) \subseteq A \times (B \cap C)$.

Let $(x, y) \in (A \times B) \cap (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \in A \times C$ (definition of intersection).
From $(x, y) \in A \times B$, we know that $x \in A$ and $y \in B$ (definition of Cartesian product). Similarly, from $(x, y) \in A \times C$, we know that $x \in A$ and $y \in C$ (definition of Cartesian product).
Therefore, $x \in A$ and $y \in B \cap C$ (definition of intersection). Hence $(x, y) \in A \times (B \cap C)$ (definition of Cartesian product).

$\square$

Ex. 6.2.31, 35. Use the element method for proving a set equals the empty set to prove each of the following statements. Assume that all sets are subsets of a universal set U.

31. For all sets A, B, and C,

$$\text{if } A \subseteq B \text{ and } B \cap C = \varnothing, \text{ then } A \cap C = \varnothing.$$

35. For all sets A, B, and C,

$$\text{if } B \cap C \subseteq A, \text{ then } (C - A) \cap (B - A) = \varnothing.$$

SOLUTION . **31.** Let A, B and C be any sets such that $A \subseteq B$ and $B \cap C = \varnothing$.

Suppose not, i.e. suppose $A \cap C \neq \varnothing$. Then $\exists x \in A \cap C$. Then $x \in A$ and $x \in C$ (definition of intersection).

Since $x \in A$ and $A \subseteq B$ and we must have $x \in B$. Therefore, $x \in B \cap C$, which contradicts $B \cap C$ is empty.

35. Let A, B and C be any sets such that $B \cap C \subseteq A$.

Suppose not, i.e., suppose $(C - A) \cap (B - A) \neq \varnothing$. Then $\exists x \in (C - A) \cap (B - A)$. Then $x \in C - A$ and $x \in B - A$ (definition of intersection).

From $x \in C - A$, we know that $x \in C$ and $x \notin A$, and similarly, from $x \in B - A$, we know that $x \in B$ and $x \notin A$ (definition of set difference).

From $x \in B$ and $x \in C$, we know that $x \in B \cap C$. Since $B \cap C \subseteq A$, we have $x \in A$. This contradicts $x \notin A$.

$\square$

Ex. 6.3.20. Prove the statement if it is true and find a counterexample if it is false. Assume all sets are subsets of a universal set U.

For all sets A and B, $\mathscr{P}(A \cap B) = \mathscr{P}(A) \cap \mathscr{P}(B)$.

SOLUTION . The statement is true.

Proof:

1. $\underline{\mathscr{P}(A \cap B) \subseteq \mathscr{P}(A) \cap \mathscr{P}(B).}$
   Let $X \in \mathscr{P}(A \cap B)$. Then $X \subseteq A \cap B$ (by definition of power set), that is, every element of X is also an element of both A and B.
   Therefore $X \subseteq A$ and $X \subseteq B$ (by definition of subset).
   Therefore $X \in \mathscr{P}(A)$ and $X \in \mathscr{P}(B)$ (by definition of power set).
   Therefore $X \in \mathscr{P}(A) \cap \mathscr{P}(B)$ (by definition of intersection).
2. $\underline{\mathscr{P}(A) \cap \mathscr{P}(B) \subseteq \mathscr{P}(A \cap B).}$
   Let $X \in \mathscr{P}(A) \cap \mathscr{P}(B)$.
   Then $X \in \mathscr{P}(A)$ and $X \in \mathscr{P}(B)$ (by definition of intersection).
   Therefore $X \subseteq A$ and $X \subseteq B$ (by definition of power set),
   Note that every element of X is also an element of both A and B. Therefore $X \subseteq A \cap B$ (by definition of intersection).

By Steps 1 and 2, we conclude that $\mathscr{P}(A \cap B) = \mathscr{P}(A) \cap \mathscr{P}(B)$. $\square$

Ex. 6.3.34. Construct an algebraic proof for the given statement. Cite a property from Theorem 6.2.2 for each step.

For all sets A and B, $A - (A - B) = A \cap B$.

SOLUTION . For some universal set U containing A and B,

$$\begin{aligned}
A - (A - B) &= A - (A \cap B^c) && \text{by the set difference law} \\
&= A \cap (A \cap B^c)^c && \text{by the set difference law} \\
&= A \cap (A^c \cup (B^c)^c) && \text{by De Morgan's law} \\
&= A \cap (A^c \cup B) && \text{by the double complement law} \\
&= (A \cap A^c) \cup (A \cap B) && \text{by the distributive law} \\
&= \varnothing \cup (A \cap B) && \text{by the complement law for } \cap \\
&= (A \cap B) \cup \varnothing && \text{by the commutative law} \\
&= A \cap B && \text{by the identity law for } \cup.
\end{aligned}$$

□

Ex. 6.3.42. Simplify the given expression. Cite a property from Theorem 6.2.2 for each step.

$$(A - (A \cap B)) \cap (B - (A \cap B)).$$

SOLUTION . For some universal set U containing A and B,

$$\begin{aligned}
&(A - (A \cap B)) \cap (B - (A \cap B)) \\
&= (A \cap (A \cap B)^c) \cap (B \cap (A \cap B)^c) && \text{by the set difference law} \\
&= (A \cap (A^c \cup B^c)) \cap (B \cap (A^c \cup B^c)) && \text{by De Morgan's law} \\
&= ((A \cap A^c) \cup (A \cap B^c)) \cap ((B \cap A^c) \cup (B \cap B^c)) && \text{by the distributive law} \\
&= (\varnothing \cup (A \cap B^c)) \cap ((B \cap A^c) \cup \varnothing) && \text{by the complement law for } \cap \\
&= ((A \cap B^c) \cup \varnothing) \cap ((B \cap A^c) \cup \varnothing) && \text{by the commutative law} \\
&= (A \cap B^c) \cap (B \cap A^c) && \text{by the identity law for } \cup \\
&= A \cap (B^c \cap (B \cap A^c)) && \text{by the associative law for } \cap \\
&= A \cap ((B^c \cap B) \cap A^c) && \text{by the associative law for } \cap \\
&= A \cap (\varnothing \cap A^c) && \text{by the complement law for } \cap \\
&= A \cap (A^c \cap \varnothing) && \text{by the commutative law for } \cap \\
&= A \cap \varnothing && \text{by the universal bound law for } \cap \\
&= \varnothing && \text{by the universal bound law for } \cap.
\end{aligned}$$

Alternative solution:

$$
\begin{aligned}
&(A-(A\cap B))\cap(B-(A\cap B)) && \\
&= [A\cap(A\cap B)^c]\cap[B\cap(A\cap B)^c] && \text{by the set difference law} \\
&= \{[A\cap(A\cap B)^c]\cap B\}\cap(A\cap B)^c && \text{by the associative law for } \cap \\
&= \{A\cap[(A\cap B)^c\cap B]\}\cap(A\cap B)^c && \text{by the associative law for } \cap \\
&= \{A\cap[B\cap(A\cap B)^c]\}\cap(A\cap B)^c && \text{by the commutative law for } \cap \\
&= \{[A\cap B]\cap(A\cap B)^c\}\cap(A\cap B)^c && \text{by the associative law for } \cap \\
&= [A\cap B]\cap\{(A\cap B)^c\cap(A\cap B)^c\} && \text{by the associative for } \cap \\
&= (A\cap B)\cap(A\cap B)^c && \text{by the idempotent law for } \cap \\
&= \varnothing && \text{by the complement law for } \cap.
\end{aligned}
$$

$\square$